RAMCO INSTITUTE OF TECHNOLOGY

DEPARTMENT OF MECHANICAL ENGINEERING

PRESENTS

RIT

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VISION MISSION STATEMENTS

RAMCO INSTITUTE OF TECHNOLOGY
DEPARTMENT OF MECHANICAL ENGINEERING

COLLEGE VISION

To evolve as an Institute of international repute in offering high-quality technical education, Research and extension programmes in order to create knowledgeable, professionally competent and skilled Engineers and Technologists capable of working in multi-disciplinary environment to cater to the societal needs.

COLLEGE MISSION

To accomplish its unique vision, the Institute has a far-reaching mission that aims:

- To offer higher education in Engineering and Technology with highest level of quality, Professionalism and ethical standards
- To equip the students with up-to-date knowledge in cutting-edge technologies, wisdom, creativity and passion for innovation, and life-long learning skills
- To constantly motivate and involve the students and faculty members in the education process for continuously improving their performance to achieve excellence.

QUALITY POLICY

RIT aims to impart Quality Education in Engineering and Technology through an effective teaching-learning process, up-gradation of facilities and human resources, collaborating with Industry for promoting training and placement, research and consultancy activities with a commitment to continual improvement of Quality Management System.

DEPARTMENT VISION

Recognized nationally for its quality education, research and extension programmes in Mechanical Engineering.

DEPARTMENT MISSION

- Producing technically competent and socially responsible mechanical engineers by imparting knowledge of cutting edge technologies and contemporary issues.
- Encouraging students for active participation in Co-curricular and Extra-curricular activities.
- Motivate students to pursue higher studies and research.
- Establish centres of excellence in promising fields of Mechanical Engineering.
- Develop linkages with industries, R&D organizations and leading educational institutions in India and abroad for achieving excellence in teaching, research and consultancy activities.

PROGRAM EDUCATIONAL OBJECTIVES (PEOs)

- Graduates will be able to apply the concept of mathematics, science and engineering to solve problems in mechanical engineering/related fields and acquire professional competence.
- Graduates will demonstrate professional development by pursing higher education and professional certification in the areas of mechanical engineering.
- Graduates will function both independently and as a member of a team in accomplishment of project with social and ethical responsibility.

PROGRAM SPECIFIC OUTCOMES (PSOs)

- Design components of products and develop systems using CAD/CAE tools.
- Provide manufacturing solutions, including material, process(es), process plans and inspect products and their components.
- Design and analyse heat power (thermal) devices/system.
- Develop simulation models of thermal and manufacturing systems using Computer aided analysis tools.

EDITORIAL MESSAGE

Welcome to the first edition of RIT Techno-Mechanic Issue of the annual magazine of 2019-2020. Department of Mechanical Engineering, Ramco Institute of Technology, Rajapalayam. We are really proud and exuberant to acclaim that we are ready with all new hopes and hues to bring out the first issue of technical contents, which is going to surely unfold the unraveled world of the most unforgettable and precious moments of the department.

The magazine is to be viewed as a launch pad for the students creative urges to blossom naturally. As the saying goes, mind like parachute works best when opened. This humble initiative is to set the budding minds free allowing them to roam fee in the realm of imagination and experience to create a world of beauty in words.

The enthusiastic write ups of our young creators are indubitably sufficient to hold the interest and admiration of the readers.

This souvenir is indeed a pious attempt to make our budding talents give shape to their creativity and learn the art of being aware because I believe that success depends upon our power to perceive, the power to observe and the power to explore. We are sure that the positive attitude, hard work, sustained efforts and innovative ideas exhibited by our young buddies will surely stir the mind of the readers and take them to the surreal world of unalloyed joy and pleasure. We have put in relentless efforts to bring excellence to this treasure trove.

It gives me immense pleasure to ensure that this magazine has successfully accomplished its objective. The reflection of the students' creativity and achievements is the epitome of the magazine. Students have put forth their ideas and thoughts that are too deep to be expressed and too strong to be suppressed.

CONTENTS

STUDENT ARTICLES

FACULTY ARTICLES

RAPID RISE IN MONETIZATION OF IOT'S ATTACK

By S.Deepak Krishna, II Year Mechanical

/deepakkrishna2000

An investigation by Trend Micro into the dark dealings of the cyber underground has found a rapid increase in the monetization of IoT attacks.

In a report released today, the global security software company revealed that forums across Russian, Portuguese, English, Arabic, and Spanish language-based markets are all brimming with chatter of how to compromise devices and then exploit them for profit. Routers and IP cameras were the most prominently discussed devices.

Financially driven attacks were found to be most prominent in the Russian and Portuguese markets, which are also the most criminally sophisticated. In these forums, cyber-criminal activity is focused on selling access to compromised devices—mainly routers, webcams, and printers--so they can be leveraged for attacks.

The greatest threat is posed to consumer IoT devices, but businesses are also at risk as hackers are increasingly wising up to the possibility of compromising connected industrial machinery to launch digital extortion attacks.

In light of their findings, researchers at Trend Micro have made four sagacious predictions that reach varying levels of doom. The first is that the move from 4G to 5G will work very much in the hackers' favor, opening up more avenues for exploitation than they've ever had before.

The second is that attacks on VR devices and cryptocurrency mining kits are going to take off big time, with more advanced threats like low-level rootkits and firmware infections on the horizon as well.

A third prophetic warning is that digital extortion attacks are going to rise as programmable logic controllers (PLCs) and HMIs are increasingly found online. Manufacturers should be cognizant that their machinery is at risk of being hijacked and their production lines halted by hackers chasing big-dollar ransoms.

Finally, the company's team of security experts reckon that attacks on routers are going to evolve entirely as ISPs become better acquainted with tactics that take advantage of DNS settings.

"We've lifted the lid on the IoT threat landscape to find that cyber-criminals are well on their way to creating a thriving marketplace for certain IoT-based attacks and services," said Steve Quane, executive vice president of network defense and hybrid cloud security for Trend Micro.

"Criminals follow the money—always. Enterprises must be ready to protect their Industry 4.0 environments."

Tomy hand

It is a 3D printed model of 3 finger hand which is being operated by single servo motor. It consist of single hexagon to which there is a three finger attached at an angle between two adjacent fingers are 120 degree. It was inspired from the movie IRON MAN, in which Robert Downey Jr would prepare his iron man suit with the aid of his own prepared robotic arm which would assist him in taking required things. When the power is given to the servo motor, it rotates the rotor which is connected to the three fingers through c clamp, so that the fingers could pick the object

Designed and fabricated by
M.Akilan III year Mech
S.Nithish IV Year Mech

PENCIL ART-I

BY

V Deepak Kumar

3rd yr Mech

deepak_17

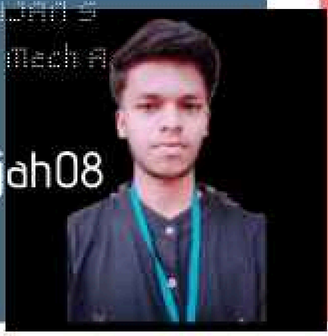
Program to determine the time of death

coded by

MAHARAJAN S

3rd year Mech A

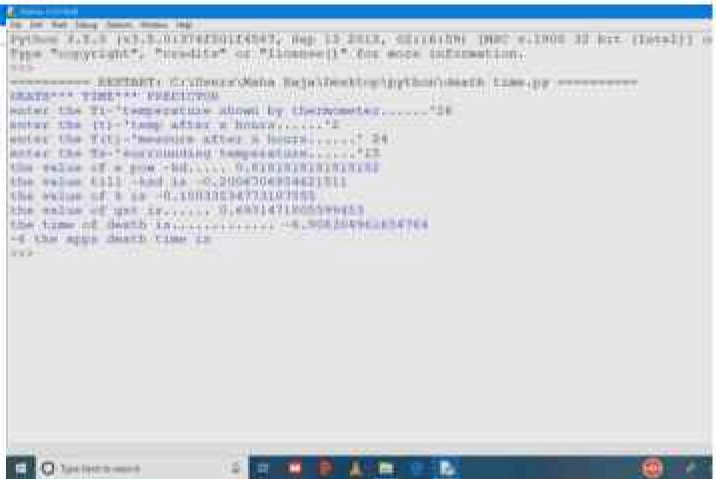
raa_jah08



This is a easy method for finding the death time of a person by checking just the temperature of the dead body. When you see the dead body measure the temperature of the body at that instant. Then wait for some hours(normally 2 hours) and again measure the temperature. Note the surrounding temperature too.

Then feed all the values to the program it will generate the time of death of the person You will get answer only in negative order The negative sign denotes that you have to subtract the answer from the current time Thus the death time of the person is calculated without any difficulty

```
print("DEATH*** TIME*** PREDICTOR")
c=float(input("enter the Ti-'temperature shown by thermometer.......'"))
d=float(input("enter the (t)-'temp after x hours......'"))
a=float(input("enter the T(t)-'measure after x hours......' "))
b=float(input("enter the Ts-'surrounding temperature......'"))
e=((a-b)/(c-b))
print("the value of e pow -kd.....",e)
import math
f=math.log(e)
print("the value till -kxd is",f)
g=f/d
print("the value of k is",g)
a=37
h=((a-b)/(c-b))
j=math.log(h)
print("the value of qxt is......",j)
k=j/g
print("the time of death is..............",k)
import math
print(math.ceil(k),"the appx death time is")
```



POEM-1

கவலை வேண்டாம்

நீ வீழ்ந்தாலும் விதையாகிறாய்

உரமிட்டு உரையாடி பேண பெற்றோர் உள்ளனர்

நீரூற்றி வளர்க்க என்றும் நிழலாய் தோழன் இருக்கிறான்

இவ்விரண்டும் உள்ளவரை உன் தோல்விகள் கூட துரத்த பயம் கொள்கிறது.....

(953616114018)
DHINESH RAJA A
dhi_ra_002

அரசர்களின் அண்ணம் கல்வி !
ஆசான்களின் ஆசனம் கல்வி !
இனிமையின் இலக்கணம் கல்வி !
ஈகைபண்பு ஈன்றும் கல்வி !
உவமை உருக்கும் கல்வி !
ஊழலை உளற்றும் கல்வி !
எவற்றுக்கும் எளிது கல்வி !
ஏற்றங்களின் ஏணி கல்வி !
ஐம்பொன் நான்திக் கல்வி ! ஒவ்வாமை ஒளித்தது கல்வி !
ஓட்டம்ஆதி முதல்அறுபது கல்வி !
....
இத்தகைய கல்வியை எங்களுக்கு வரமளித்த ஆசிரிய பெருமக்களுக்கு
ஆசிரியர் தின நல்வாழ்த்துக்கள் !!!!!

நன்றி சொல்வது ஒரு கலை

அதனை மறப்பது பெரும் பிழை

தாய், தந்தைக்கான நன்றி என்பது கடமை

ஆசிரியருக்கான நன்றி என்பது பெருமை

மருத்துவருக்கான நன்றி என்பது செழுமை

ஆனால்

ஓட்டுநருக்கான நன்றி என்பது அருமை

இன்றிலிருந்தாவது சொல்வோம் நம் இன்றியமையா
பயணத்தின் பாதுகாவலனுக்கு நன்றி என்று

POEM

J.BASKAR

IV YEAR MECHANICAL

Baskar_J

கனவுகளோடு கண்கள்

அறிவை விற்க இடம் தேடி
ஆகாய அளவின் ஆசைகளோடு
இறுக்கம் கொண்ட மனதோடு
ஈன்ற எம்மக்களின் வலியோடு
உருக்கம் மறந்த விழியோடு
ஊனை உருக்கும் சுமையோடு
எல் அளவு வெற்றியாவது கிடைத்துவிடாதா
ஏற்றத்தை வாழ்கை காட்டிவிடாதா என்று
ஒருநாள் அவ்வொருநாள் மகிழ்விக்கு
ஓடி ஓய்ந்து தேய்ந்த கால்களோடு
ஔவை கூறிய சான்றோனாய்
எதிர்கால வாழ்வை நோக்கி காத்திருக்கும் எனது ....
கனவுகளோடு கண்கள் ...

இனிய புத்தாண்டு நல்வாழ்த்துக்கள்

அன்பு உன் உணர்வுகளிலிருந்து
ஆசை உன் கனவுகளிலிருந்து
இன்பம் உன் மனதிலிருந்து
ஈகை உன் கைகளிலிருந்து
உண்மை உன்சொற்களிலிருந்து
ஊக்கம் உன் வலிகளிலிருந்து
எளிமை உன் குணத்திலிருந்து
ஏற்றம் உன் செல்வங்களிலிருந்து
ஒற்றுமை உன் உறவுகளிலிருந்து
தொடங்கட்டும்
இம்மாற்றங்களோடு தொடங்கட்டும்

உன் புத்தாண்டு.........

தேடல்....

அறிவியலில் வளர்ச்சியை தேடு
ஆன்மீகத்தில் அமைதியை தேடு
இல்லறத்தில் ஒற்றுமையை தேடு
ஈமத்தில் மனிதர்தளை தேடு
உழைப்பை உன்னிடம் தேடு
ஊணில் ஆரோக்கியத்தை தேடு
எண்களை உதவிகளில் தேடு
ஏழ்மையை தவறுகளில் தேடு
ஒழுக்கத்தை பழக்கத்தில் தேடு
ஓய்வை உன் வாரிசுகளில் தேடு......

- கவிஞர் சாரா பா

PENCIL ART-II

HE DID IT!

R SUBHASH
IV YEAR MECHANICAL
/subhash_aved

A Day In My Life

S.NITHISH
IV YEAR MECH
/Nithish SNP

INDUSTRY 4.0 USING INTERNET OF THINGS (IOT)

INTRODUCTION

Transformation in Manufacturing Industry is happening by digitizing of manufacturing. This transition is being called as Industry 4.0 which is the fourth revolution occurred in manufacturing industry.

Whereas in Industry 3.0 adaptation of computers and automation enhanced the smart and autonomous system fuelled by data and machine learning. In Industry 4.0, the machines were allowed to communicate with each other and make decisions without humans' involvement. It is a combination of cyber-physical systems, the Internet of Things and the Internet of Systems make Industry 4.0 possible and the smart factory a reality. With the support of smart machines that keep getting smarter as they get access to more data, our factories will become more efficient and increases production with less wasteful.

CASE STUDIES

- By using the data from sensors in its equipment, an African gold mine identified a problem with the oxygen levels during leaching. Once fixed, they were able to increase their yield by 3.7%, which saved them $20 million annually.
- The MG Hector is India's first car that will offer connectivity on the go and the brain behind it is the revolutionary iSMART Next Gen technology. It combines hardware, software, connectivity, services and applications to make your driving experience easier, smoother, and smarter.
- Boasting the most-advanced Industry 4.0 standards, the futuristic facility merges virtual worlds and live production by integrating robotics and machine-to-machine collaboration with Lamborghini's s killed production workers. Each vehicle moves along the production floor via automatic guided vehicles (AGVs) that autonomously transport each car to the appropriate work island.

LIVE CASE STUDIES IN VIRUDHUNAGAR DISTRICT

Recently,I had an opportunity to visit Sree Ayyanar Spinning and Weaving Mills Limited, Mallanginar which is an Pioneer Asia group of companies certified by ISO 9001:2000. Where they use 58,000 spindles (6000 spindles for weaving and remaining for spinning of cotton) in their mallanginar plant. They found very difficult is collection of data such as

1. Breakage of cotton spinning
2. When the breakage was attended and completed?
3. What was the idle time of spindle?
4. What was the cost incurred by machine due to idle time?

The above said problems caused a very high loss in productivity and increased in wastage. Hence the technical team came out with an idea of implementation of IoT in their spinning and weaving plant. After implementing they had an increase in productivity and less wastage.

Author

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/praburam-ganesamoorthy-78001457/

References

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AI

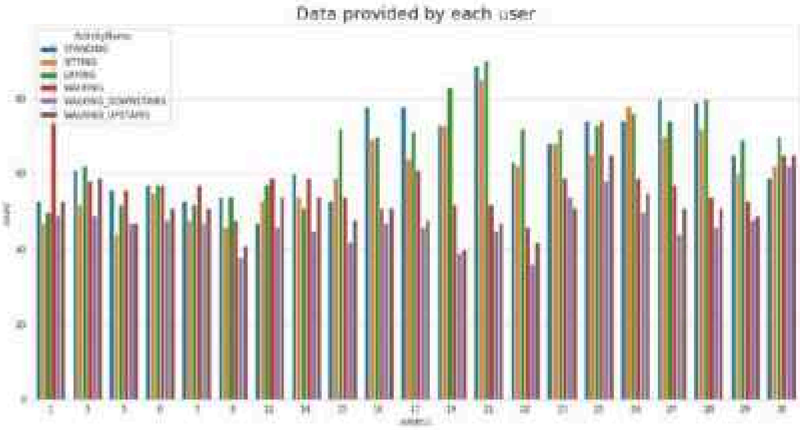
Human Activity Recognition using Deep Learning Models

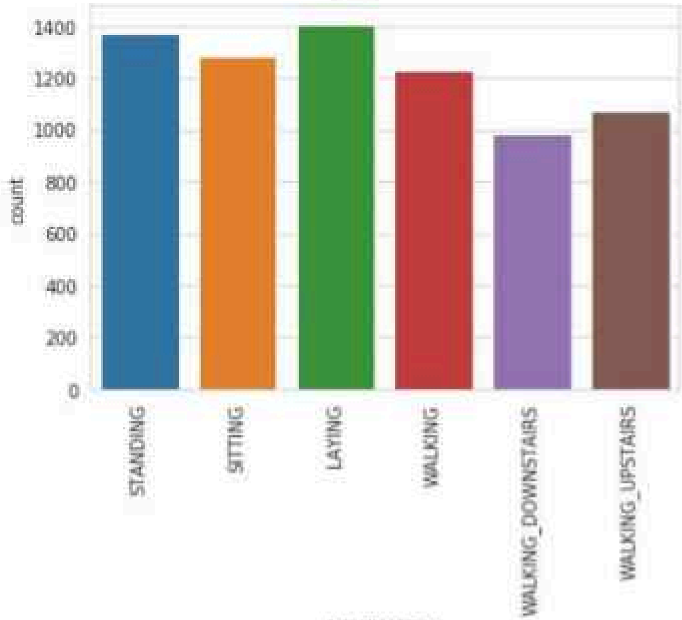
Mr.S.Valai Ganesh
Assistant Professor
Mechanical Engineering
/valaiganesh

Human activity recognition (HAR) is a latest technology that can be used for identifying human activities through computer and smart phone systems. A standard human activity recognition dataset is the `Activity Recognition Using Smart Phones Dataset' available from internet. Input signals can be taken from different types of detecting devices such as audio sensors, image sensors, barometers, gyroscopes and accelerometers. The rapid development of communication systems between human-computer and human-smart phones leads to detect activities of humans in every aspect. More importantly, the recent introduction of GPU and deep learning algorithms has made human activity recognition widely used in athletic competition, smart home automation and health care for the old peoples.

Long Short Term Memory is proposed to detect the human activities based on Human Activity Recognition (HAR) dataset. The data is recorded with the help of sensors (accelerometer and Gyroscope) in smart phone. This dataset is collected from 30 persons, performing different activities with a smart phone to their waists. The testing of the model is evaluated with respect to accuracy and efficiency. The designed activity recognition system can be manipulated in other activities like predicting abnormal human actions, disease by human actions,etc.

There are two packages are added with LSTM. One is Hyperopt and Hyperas. Hyperopt is a python library for serial and parallel optimization over difficult search spaces, which may include real valued, discrete, and conditional dimensions of data. Hyperas is Simple wrapper for hyperopt to do convenient hyperparameter optimization for keras models. Softmax activation function is used here to predict six different classes.





The prepared model is classified in to 70:30 ratio. That means 70% of dataset is used for training the model and rest 30% is used for testing. The model produced good results of around 95% validation accuracy with LSTM model with hyper parameters. The same dataset is tested with conventional machine learning models and it produced around 88% validation accuracy. With the aid of AI and Deep Learning Technique we can able to identify human behaviour under abnormal situation also.

ADDITIVE MANUFACTURING (3D PRINTING)

DEFINITION

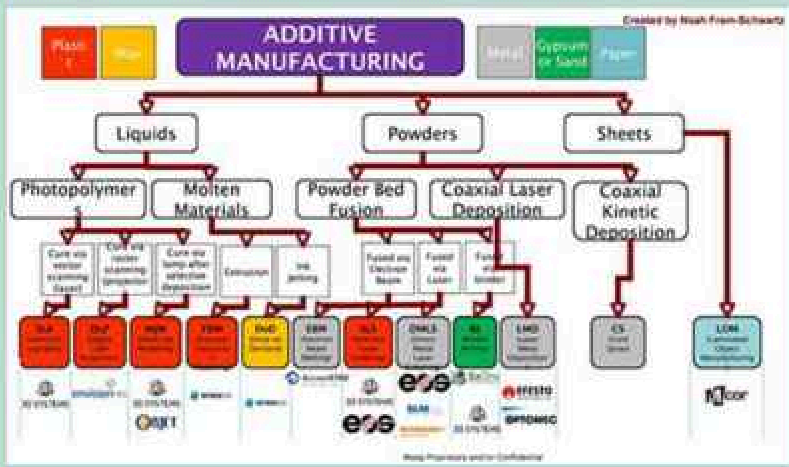
A prototype is the first or original example of something that has been or will be copied or developed; it is a model or preliminary version;
e.g.: A prototype supersonic aircraft.

An approximation of a Product (or system) or its components in some form for a definite purpose in its implementation.

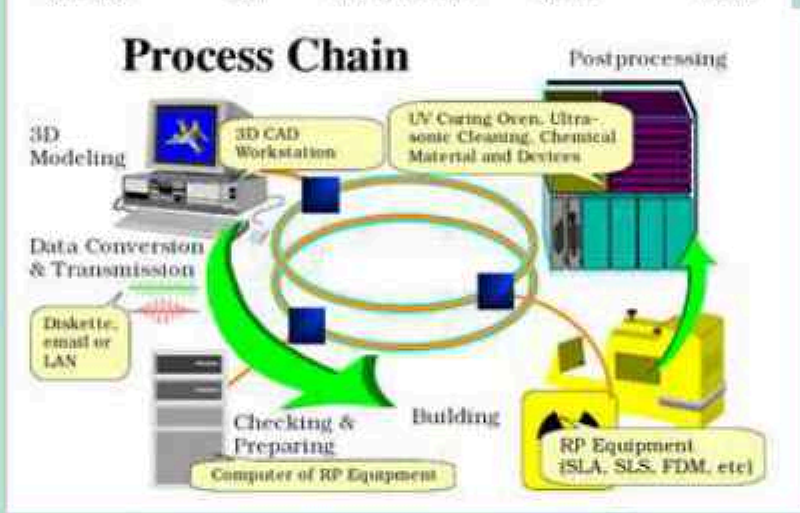
Case Study 1 – Personal Memento

3D printing in the medical field

CLASSIFICATIONS



MAJOR ASPECTS



By

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Mr.R.Prabhakaran,
Assistant Professor/Mechanical
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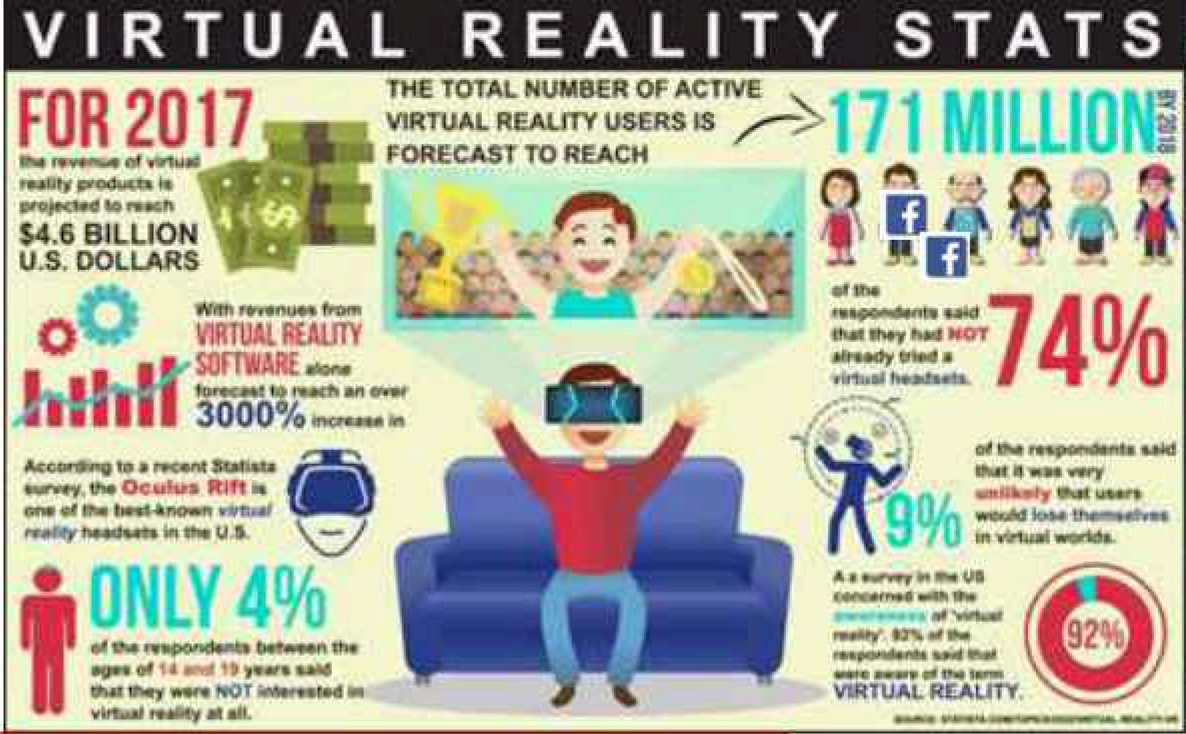

AUGMENTED AND VIRTUAL REALITY

By

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/Prabhakaran.MIzzY

Augmented reality (AR) adds digital elements to a live view often by using the camera on a smartphone.

Examples of augmented reality experiences include Snapchat lenses and the game Pokemon Go.

Virtual reality (VR) implies a complete immersion experience that shuts out the physical world. Using VR devices such as HTC Vive, Oculus Rift or Google Cardboard, users can be transported into a number of real-world and imagined environments such as the middle of a squawking penguin colony or even the back of a dragon.

There are apps available for or being researched for AR in nearly every industrial sector including:

- Archaeology, Art, Architecture
- Commerce, Office
- Construction, Industrial Design
- Education, Translation
- Emergency Management, Disaster Recovery, Medical and Search and Rescue
- Games, Sports, Entertainment, Tourism
- Military
- Navigation

BATTERY MANAGEMENT SYSTEM: THE KEY TO ELECTRIC VEHICLE ADOPTION

Article by Dr.V.Sivakumar, Assistant Professor(Senior Grade)/Mechanical

R6 /Sivakumar_V4

The Government of India set an ambitious target of 6-7 million electric vehicles (EVs) by the end of 2020 under the National Electric Mobility Mission (NEMM) Plan 2013.

This is an effort towards ensuring EV manufacturing leadership and national fuel security. Successful adoption of EVs will require high-performance batteries (high energy and power density with long cycle life) and an advanced Battery Management System (BMS). BMS,an electronic system, protects a battery from over-charging/over-discharging which ensures the safe operation of EVs. In the context of a fast growing EV market in India, it is important to develop a comprehensive and mature BMS programme.

The latest research and development trends indicate that the next generation of high-energy lithium-ion battery (LIB) systems will soon be available for electric transportation.

However, high costs and low awareness, especially related to the safety features and drive range provided by EV batteries, are considered to be the key barriers in EV adoption. Among all the components of EVs, the battery packs which include a battery and BMS, alone account for 40% of the total cost of a vehicle. Currently, Indian EV manufacturers like Ashok Leyland and Mahindra and Mahindra depend on imported LIBs and BMS, which result in high costs. As per the Global and China Power Battery Management System Industry Report, 2016-2020, the market size of BMSs would reach USD 7.25 billion by 2022 from the USD 1.98 billion mark as reported in 2015. This is estimated at a compound annual growth rate of 20.5%. North America held the largest share of the BMS market in 2015, followed by Europe, Asia Pacific and the rest of the world. This is estimated at a compound annual growth rate of 20.5%. North America held the largest share of the BMS market in 2015, followed by Europe, Asia Pacific and the rest of the world. Aiming for large-scale EV deployment in India by 2030 under NEMM, Indian EV manufacturers can import different components of EV along with the BMS.

In addition, global EV manufacturers can set-up their integrated plant operations in India in the longer term.

Most electric bus manufacturers develop their own BMSs. For instance BYD, Youtong, Proterra and EBUSCO make "intelligent BMS" for electric buses.

On the other hand, Wuzhoulong Motors outsource BMS manufacturing to the largest lithium iron phosphate battery manufacturer company, Optimum Nano. In India, KPIT Technologies introduced the first smart electric bus in 2015, aligned with the government's Make in India initiative. KPIT also develops BMSs in-house. Ashok Leyland introduced electric buses in India in 2016 with batteries and BMS imported from France. An advanced BMS can be developed in India through carefully structured R&D. Under the Make in India initiative, in-house process manufacturing capability of LIBs along with BMS will help in reducing the total cost of EVs

RECENT TRENDS IN SOLAR ENERGY

Mr.K.Vigneshwaran
Assistant Professor
Mechanical Engineering
/wiki_clickz

In the past decades, Solar power had been achieving milestones in Solar Energy storage, Increase in Solar panel efficiency and saw new models in its design. Generally, Solar technology has been classified into two main types: 1) Solar PV and 2) Solar thermal.

What's new in 2019?

The following are the some of the emerging Solar technologies, achievements and research areas in 2019

Solar powered Roads: These designed roads have the ability to generate clean energy and also have LED bulbs that can light roads at night. The sidewalks along Route 66, America's historic interstate highway, were chosen as the testing location for solar-powered pavement technology.

PV integration: Solar Panels + Inverter: In solar panels the concept of efficiency refers to the rate at which the silicon cells that converts light energy to electricity. For Installers, installation of both solar panel and inverters separately can sum up more time. In order to reduce this time and difficulty, inverters are fixed back of the solar panels itself.

Wearable Solar: Now, the Solar panels has been in our clothes too. Even though we knew about Solar watches and other gadgets. It found its new innovation in textiles. Tiny solar panels can now be stitched into the fabric of clothing, window urtains and dynamic consumer clean technology like heated car seats. It is typically made of hard plastic material.

Solar thermal fuel (STF): Solar thermal fuel appears to be the best alternative storage than solar batteries economically. The solid state STF application that could be implemented in windows, windshields, car tops, and other surfaces exposed to sunlight. This technology and process behind STFs is comparable to a typical battery. The STF use sunlight energy, store it as a charge and then release it when prompted. The issue with storing solar as heat is that heat will always dissipate over time, which is why it is crucial that solar storage technology can charge energy rather than capture heat.

MISCONCEPTIONS ABOUT SOLAR ENERGY IN ELECTRICITY GENERATION

Efficiency

Efficiency of photosynthesis in most plants is less than 1%. The efficiency of coal electricity is about 8-10%. The efficiency of commercial solar module available today is about 15-18%. We do not buy a TV, car, fridge, etc by considering only the efficiency. It is the cost of electricity that matters, not just efficiency.

Space

With current Solar PV technology, one can generate about 100 units of electricity/month from 10 Sq.Meter panel/rooftop area, including space for maintenance. An average Indian consumes about 100 units/month. Even upper-middle class, will consume 300-400 units/month, i.e. 30-40 Sq. meter. Every house has enough space (except multi-story building that constitutes a very small percentage of population). Even if we want to generate 100% of electricity in India from Solar, we need only 0.2% surface area.

Expenses

The per unit grid electricity cost of domestic users is Rs. 5-6 /unit and for industrial users Rs. 7-8 /unit. The cost of solar PV electricity without storage is Rs. 3-4/unit, including capital cost and about Rs. 5-7 /unit including storage. A lot of businesses are willing to sell electricity without requiring users to make any investment.

Maintenance

It requires significant maintenance - The solar PV module generates electricity on exposure to sunlight without moving any parts, unlike coal power plants. In the absence of any moving parts, the only maintenance required is to clean the dust. As a result, the life of PV modules is very long. As per definition, PV modules degrades only by 20% in its lifetime of 25 years, i.e. even after 25 years, a good PV module will perform 80% of its initial capacity. In terms of electricity storage, with every solar PV system, you do not need battery storage. Under net metering, one can connect the PV system to the electricity grid and the grid becomes battery storage. For a standalone PV system, one needs batteries. There are batteries like sealed maintenance free, which requires nearly no maintenance.

Seasonal Effect

The solar panel output depends on the input. As long as some light is falling on the panel, there will be some output. Most part in the country, except the coastal areas, 50-80% radiation falls during the rainy season compared to normal days, hence solar panel will still produce 50-80%. On full overcast days, also there will be about 30-40% generation.

By

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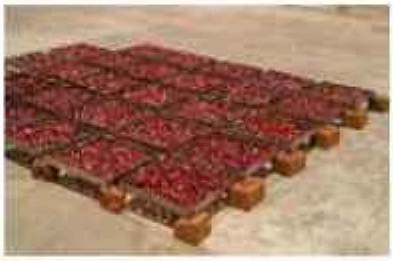
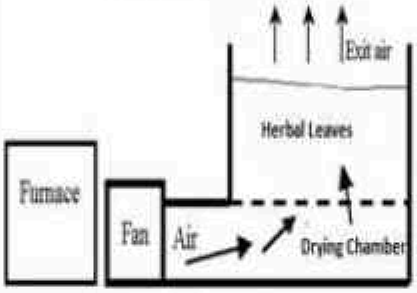
DRYING TECHNIQUES IN HERBAL PROCESS INDUSTRIES



Open sun drying - Leaves are directly exposed to solar radiation

Shade drying – Drying by UV free ambient air

Wind Drying - Bunch of leaves drying with UV free ambient air



Fixed Bed Drying – The processed hot air passes through the herbal leaves

Fluidized Bed Drying – Leaves are suspended in drying chamber by balancing the weight of leaves and velocity of hot air

Heat pump drying – The dehumidied air attained by cooling & dehumidification and by heating is circulated continuously

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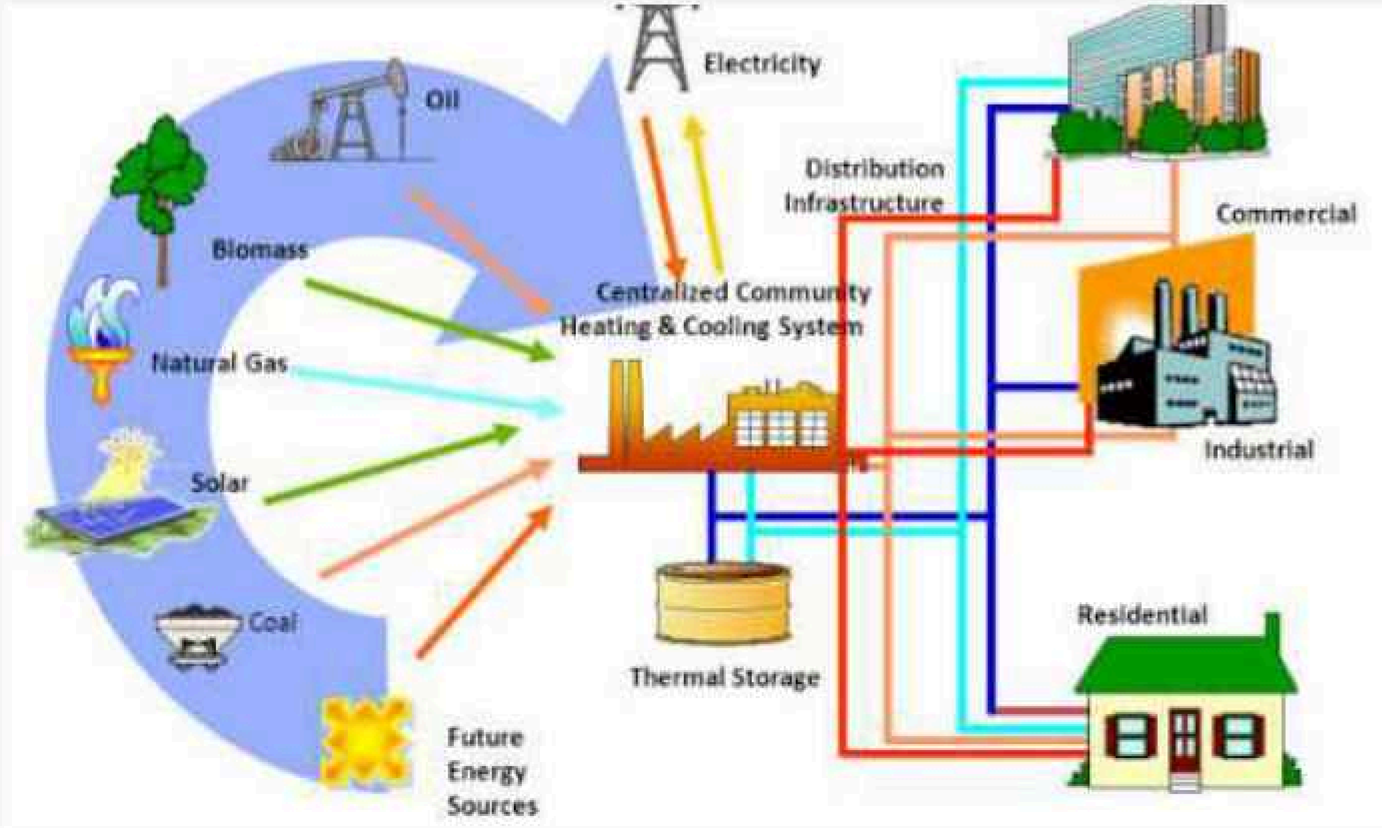
Drying is one of the most important methods to preserve the Herbal leaves for a long period of time without fungal growth and for making medicine in powder form used in herbal processing industries. The drying techniques are classified as given below

- Thermal Drying
  - Natural Drying
  - Forced Convection Drying
- Chemical Drying
  - Glycerin drying
  - Silica sand drying
  - Calcium chloride drying
- Special Drying
  - Press drying
  - Green house drying
  - Freeze drying
  - Vacuum oven drying
  - Microwave drying
  - Microwave vacuum drying

DISTRICT COOLING SYSTEM

DISTRICT COOLING SYSTEM (DCS) DISTRIBUTES THERMAL ENERGY IN THE FORM OF CHILLED WATER FROM A CENTRAL SOURCE TO MULTIPLE BUILDINGS THROUGH A NETWORK OF UNDERGROUND PIPES FOR USE IN SPACE COOLING. THE COOLING OR HEAT REJECTION IS USUALLY PROVIDED FROM A CENTRAL COOLING PLANT, THUS, ELIMINATING THE NEED FOR SEPARATE SYSTEMS IN INDIVIDUAL BUILDINGS.



District cooling system is known as District heating system when used for heating in colder countries. District cooling or heating systems are widely used across the globe. DCS consists of three primary components,

1. The central cooling plant.
2. The chilled distribution network.
3. The consumer system or energy transfer station.

The central plant generates chilled water with compressor drive chillers or absorption chillers. Large size centrifugal chillers with higher efficiency are usually installed to take advantage of the economies of scale. Apart from energy efficient chillers, the central plant consists of chilled water pumps, cooling towers with condenser water pumps, thermal energy storage tank, electrical power distribution system and control system for centralized automation and control of equipment.

District Cooling is a sensible, cost-effective solution for cooling in areas such as the Middle East, Europe, Asia, and South America, among others. For urban areas, it is the perfect solution for some of the most pressing issues of the modern age. It helps to reduce the demand for power and other fuels, offering sustainability. District Cooling has a significant positive environmental effect by reducing CO_2 emissions and pollution. These systems can also reduce electric consumption by an average of 50%, and it is could be 40-60% more efficient than conventional air conditioning.

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WELDING TECHNOLOGY

Welding Technology is specialized domain in the field of Mechanical engineering. Even it is traditional manufacturing method to join materials; it is always the one of the emerging technology. Most of researches are carried out in this field to improve the effectiveness in welding for various applications.

Challenges in Welding Process are

- Less spatter, to reduce clean-up time.
- Less heat input, to reduce distortion and help with thinner materials.
- Improved ability to weld dissimilar materials.
- Better arc control, to help address problems caused by poor fit-up.
- Faster travel, so we can make welds in less time (and put less heat in.)

Due to advancement in technologies and emerging new materials, there is a need in introducing new techniques in welding technology rather than follow the conventional or traditional mode of welding.

Some of the advanced welding technologies:

- Magnetic Arc Welding
- Friction and Friction Stir Welding
- Ultrasonic, Electron and Laser Beam Welding
- TIG and MIG Welding
- Cold Metal Transfer Welding

Current researchers introduce the additive manufacturing techniques in current advanced welding process. To improve the productivity in manufacturing sector, many industries (Automotive, Nuclear, and Aerospace) adopted the concept of automation in welding by using robots, material handling and conveyor systems. Welding of metals could be carried out with using robots in the hazardous environment like mining, nuclear and underwater welding.

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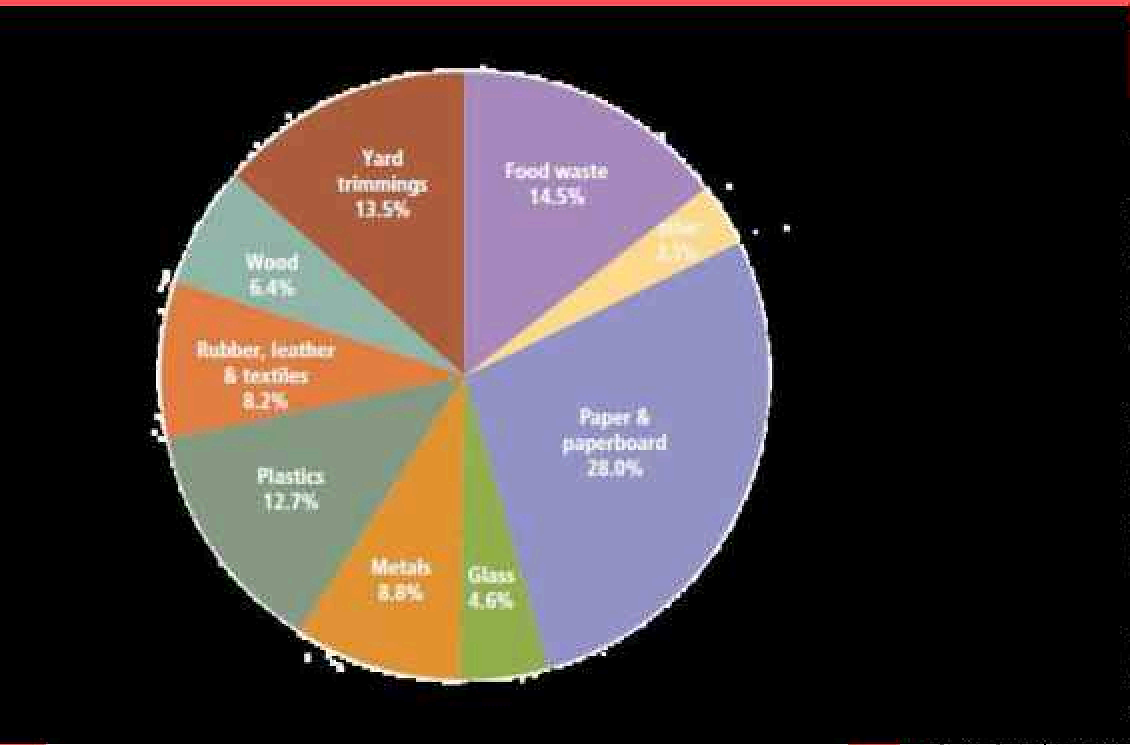


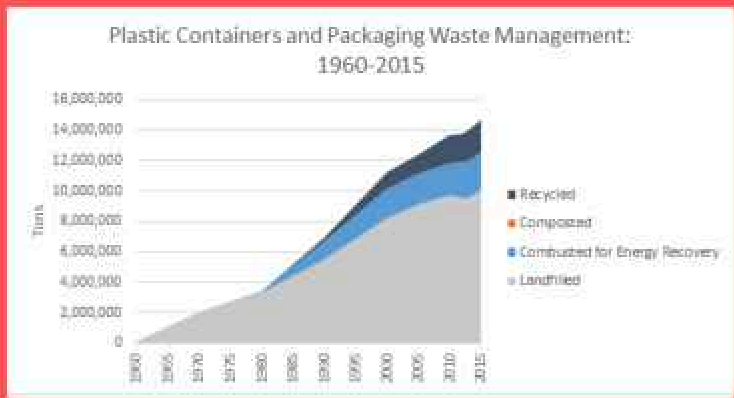
TE MANAGEMENT

...erated by the daily activities of ... These wastes are normally called ... the word 'waste' indicates that ... unwanted. But most of the ... be reused and recycled for ...ses. This waste management has ...ious issue throughout the world ...be addressed. The government ...arge to collect, transport and ...in an eco-friendly manner. The ... and quantities will differ ... the location. The content and ...the solid waste depends upon the ...e, population, social behaviour, climate changes, industrial production etc.

Municipal solid waste (MSW) is classified as commercial, domestic, industrial, institutional, demolition, construction and municipal wastes. Solid waste management becomes the difficult task due to urbanization. Quantity and composition differ due to the various countries management system which may be successful at one place and unsuccessful at another place.

At present the public and private sectors should hold the responsibility to develop the opportunities from the MSW generated by exploring the varied ways on how it can be reused and recycled instead of thinking about the problems arising from the solid waste generated.

Because there is a possibility to generate the revenue from the MSW by Segregating the recyclable waste from the total waste generated. This recyclable waste will encompass plastic, rubber, sheet metal, glass etc.



CONCLUSION

The recyclable market vendors are ready to purchase these recyclable materials. It involves the management of the activities associated with generation, storage, collection, transfer, transport, processing and disposal of solid waste that will be environmentally compatible by adopting principles of economy, aesthetics, energy and conservation.

But most of the developing countries like India they are the beginners in the recycling, reusing activities because of this reason landfill is the most adoptable one to reduce the impact of solid waste and its harmful effects.

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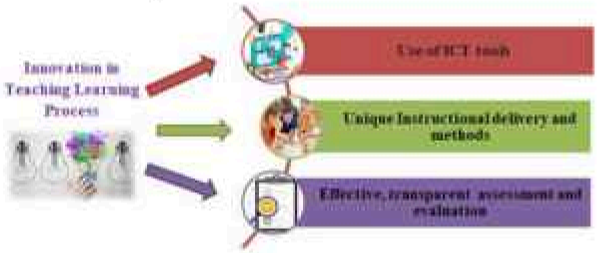
INNOVATIONS IN TEACHING LEARNING PROCESS – PEDAGOGY FOR Z-GEN STUDENTS

Practice of effective and meaningful teaching can benefit the students immensely when educators thoughtfully experiment and apply new or different pedagogical approaches. The goals of innovative practices in teaching learning process are to make the students to get insight knowledge, skills sets and behavioural approach in the course and obtain good grades in University examinations. Thus innovations in teaching learning process will help the students in attaining the graduate attributes.

Department of Mechanical Engineering at Ramco Institute of Technology categorizes Innovation in Teaching Learning process into three aspects as follows:

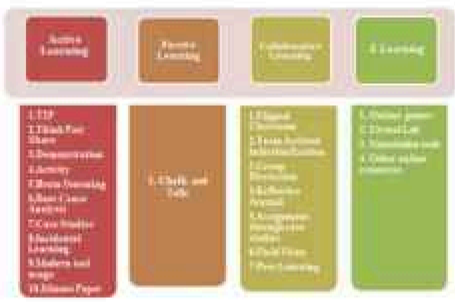


1. USE OF ICT TOOLS:

Use of Information Communication Technology (ICT) tools helps a teacher and an institution to stand unique. Use of such ICT tools helps a faculty member to engage Z-generation students in effective learning and enables learning beyond classroom. Use of ICT tools are practiced in our department through following aspects:

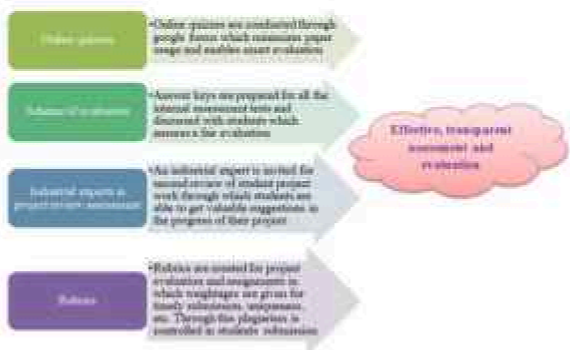
2. UNIQUE INSTRUCTIONAL DELIVERY AND METHODS:

Mix of instructional delivery and methods keeps the students to be engaged inside the classroom. We the faculty members in Department of Mechanical Engineering use four modes of instructional delivery and methods.



3. EFFECTIVE, TRANSPARENT ASSESSMENT AND EVALUATION:

Transparent evaluation in the course is brought in through use of rubrics for assignments, conduct of online quizzes through Google forms, preparation of answer key for internal tests.



Deliverables of Innovative Practices:

Innovative practices followed in our department helps the teaching community in achieving the following outcomes.

I AM MORE FASCINATING ON FINLAND EDUCATION SYSTEM. WHY?

Finnish education is high quality oriented education which consists of daycare and comprehensive education system. They mostly emphasis on lifelong learning education system. Finland's strategy for achieving educational equality and excellence is based on building a publicly funded primary school system without selecting, tracking, or streaming students during a common basic education. They provide facility based on the needs and expectations of the common people. The component of Finland education system is to provide uniform basic education for all age group, highly capable teachers and sovereignty given to schools. The curriculum is designed based on the work experience, real case studies and practical training.

The module of the course consists of several works such as lectures, guided instructions, exercises, set-book examinations, seminars and so on. The design of instruction methodology is to motivate self-learning activities. They have implemented instructor oriented methodology, where the student can do more project and team work activities.

Structure of Finland Education

Academic degrees	Vocational degrees	Typical ages
doctor	employment	
licentiate		
master	master (new)	(+2-3)
bachelor	bachelor	(+3-4)
upper secondary school (voluntary)	vocational school (voluntary)	18-19
		17-18
		16-17
comprehensive school (compulsory)		15-16
		14-15
		13-14
		12-13
		11-12
		10-11
		9-10
		8-9
		7-8
pre-school		6-7

One of the main reasons for success of this education system, the student do practical on job learning which enable many student to do their project with hands on experience and also apply their knowledge in real case studies. They also work on working life case studies and are often commissioned by enterprises.Progressions of students are monitored by personal study plans. This facilitates the planning of studies and the monitoring of progress in study support system for students and counseling. The master degrees eligibility is three years of working experience with bachelor degree. Every student should know about the assessment criteria and grading examination paper and other performance records. If the students are not satisfied with the assessment criteria, they may have freedom to do correction.

The assessment criteria may be decided for practical training by the institution itself.The scheme of evaluation is more transparent and quality basis. I like more about Finland education, because their philosophy of education is **"YOU ARE NOT TAUGHT, YOU LEARN" & "LOVE OF LEARINING"**

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GROSS DOMESTIC PRODUCT (GDP)

GDP IS THE FINAL VALUE OF THE GOODS AND SERVICES PRODUCED WITHIN THE GEOGRAPHIC BOUNDARIES OF A COUNTRY DURING A SPECIFIED PERIOD OF TIME, NORMALLY A YEAR. GDP GROWTH RATE IS AN IMPORTANT INDICATOR OF THE ECONOMIC PERFORMANCE OF A COUNTRY

Courtesy: Official websites of "The Economic Times" & "The Indian ministry of Statistics and Programme Implementation".

It can be measured by three methods, namely,

1.Output Method: This measures the monetary or market value of all the goods and services produced within the borders of the country. In order to avoid a distorted measure of GDP due to price level changes, GDP at constant prices or real GDP is computed.

2.Expenditure Method: This measures the total expenditure incurred by all entities on goods and services within the domestic boundaries of a country.

Expenditure Method = C + I + G + (X-IM)

Where

C: Consumption expenditure,

I: Investment expenditure,

G: Government spending and

(X-IM): Exports minus imports, that is, net exports.

3.Income Method: It measures the total income earned by the factors of production that is labour and capital within the domestic boundaries of a country.In India, contributions to GDP are mainly divided into 3 broad sectors – agriculture and allied services, industry and service sector. In India, GDP is measured as market prices and the base year for computation is 2011-12.

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REDUCE, RECYCLE & REUSE TODAY FOR A PLASTIC-FREE TOMORROW

Plastics once a biggest savior has now become a threat to the environment by means of plastic pollution. It can afflict land, water and air when incinerated and also affects the biodiversity of nature. From 1950 to 2018, roughly 6.3 billion tons of plastics have been produced all over the world of which only 9% got recycled and 12% got incinerated. India alone produces about 25,940 tons of plastic and more than 97,000 tons of solid waste per day. Today, plastics have become an integral part of daily life. A world without plastics cannot be imagined. But if we don't act now then, by the time we all come to our right senses, it might be too late for us.

HERE ARE SOME OF THE ALTERNATIVES THAT WE ALL CAN USE OR TRY IN PLACE OF PLASTIC PRODUCTS:

1. REUSABLE STRAWS, SPOONS, CUPS AND PLATES

INSTEAD OF USING PLASTIC STRAWS, SPOONS, CUPS AND PLATES, CARRY AND USE YOUR OWN STAINLESS-STEEL STRAWS, SPOONS, PAPER CUPS AND PAPER PLATES OR OTHER REUSABLE PRODUCTS.

2. CARRY REUSABLE BOTTLES

PLASTIC BOTTLES TAKE 450-1000 YEARS TO DECOMPOSE AND 1 MILLION PLASTIC DRINKING BOTTLES ARE PURCHASED EVERY MINUTE AROUND THE WORLD. NOW, IMAGINE THE NUMBER OF WATER BOTTLES THAT MUST HAVE BEEN DISCARDED EVERY DAY. SO, CARRY GLASS OR STAINLESS-STEEL BOTTLES INSTEAD OF PURCHASING WATER BOTTLES EVERY TIME YOU NEED DRINKING WATER.

3. CARRY BAGS FOR PURCHASE

CARRYING YOUR OWN CANVAS AND REUSABLE CARRY BAGS MADE OF COTTON, JUTE, HEMP, NYLON AND LEATHER ARE BEST ALTERNATIVES TO PLASTIC SHOPPING BAGS. MOST OF THESE BAGS ARE LIGHT WEIGHT, CUSTOMIZABLE WITH YOUR OWN DESIGNS AND EASY TO CARRY WHEREVER YOU TRAVEL.

4. PRE-CYCLE AND RECYCLE

PRE-CYCLING IS THE PRACTICE OF DETERMINING THE RIGHT NUMBER OF PRODUCTS THAT ARE ESSENTIAL AND PRODUCTS THAT COMES IN PLASTIC PACKAGES, AND FINDING WAYS TO AVOID THEM OR AT LEAST MINIMIZE THEM. IT'S ABOUT MAKING THE BEST DECISIONS FROM THE CHOICES THAT ARE AVAILABLE TO US. RECYCLING IS THE PROCESS OF TRANSFORMING WASTE PRODUCTS INTO NEW MATERIALS AND OBJECTS.

5. SPREAD AWARENESS IN SOCIETY

IT IS NOT THE DUTY OF THE GOVERNMENT OR ANY OTHER ESTABLISHMENT ALONE TO LOOK INTO THIS AFFAIR AND TURN IT OUT. EACH AND EVERY ONE OF US IS RESPONSIBLE FOR IT AND WE SHOULD FORGE ON IT UNTIL THE LAST SCRAP OF PLASTIC WASTE GETS TAKEN AWAY FROM THE SOIL OF THE WORLD.

We may not be able to win this battle in a day or two, but if we start today, tomorrow will certainly be a better day.

By
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SSB - GIVE IT A TRY

SOME GOALS ARE SO WORTHY, IT'S GLORIOUS EVEN TO FAIL

There are some Professions that make us unique with a greater sense of responsibility with new experience on every step; one among them is being a Commissioned officer in Indian defence services (Army, Navy, Air force). These officers are the leaders or brains of the troops they command. To be an officer in such forces an aspirant has to clear the Service Selection Board. The board evaluates the suitability of the candidate for becoming an officer using a standardised protocol of evaluation system which constitutes of personality, intelligence tests, and interviews. The main evaluation is that each candidate is checked for 15 officer like qualities (OLQs) and the assessors check whether the candidate's thoughts, words, actions are collinear. For that he/she will be put into various individual tasks, group tasks, psychology tests and the creamy will be filtered for the medical tests. Success rate in other competitive exams is dependent on heavy preparations and knowledge of the syllabus prescribed but to clear SSB interview, attitude and habit of the individual plays a major role. Developing a habit takes time and it is responsible for our actions and the decisions we take.

In every SSB there will be thousands of candidates appear per batch but only one or two get recommended. Some candidates crack SSB at their first attempt and some during their seventh or eighth attempt. Some prepare themselves for SSB while some clear SSB just like that as they are conscious about themselves naturally.

The candidates clearing SSB will be groomed as officers through intense training at the toughest academies (OTA, NDA, INA, and AFA) of our nation. Many youth wish to enter this career for reasons like job security,post-retirement options, pay perks and facilities for family and some may even clear SSB with these expectations but a real aspirant's preference differs. Inspiration for these candidates will be their craze for the Uniform which is a symbol pride and the ardent belief for serving our Motherland. Dr. APJ Abdul Kalam is one of the best examples of someone who became immensely successful even after failing to get selected at the SSB. What is ironical is that he became the commander of the Armed forces as the President of the country. His contributions to the country's defence sector remains unparalleled.

The current scenario is Indian army is facing a terrible shortage of officers and not shortage of men (soldiers) and at the same time majority of graduates incline for a materialistic job profile with corporate. Leave patriotism ahead, only in this career you can lead men into force at an age of 22 or 23 with your personal charisma and not through monetary incentives. This career equips one with physical and mental toughness for a much rich and empowered life. During the SSB interview, once you are screened in throughout the stay at the centre you will undergo a self analysis, your strength, weakness and you will gain a better understanding of yourself even if you don't clear the interview. One should experience SSB at least once in his/her lifetime. Do give it a try...

"If u Win,
U Can Lead!
If u Lose,
U Can Guide!"

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SPOTTED AT RIT: BLACK-SHOULDERED KITE

It is rated as least concern on the International Union for Conservation of Nature (IUCN)'s Red List of Endangered species.

Captured by

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